



C.V. Raman
Global University
ODISHA BHUBANESWAR INDIA

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ODISHA



C. V. Raman
Global University
ODISHA BHUBANESWAR INDIA

Department of Chemistry

BIOLOGY AND ENVIRONMENTAL SCIENCE (CY202)

UNIT-IV : Air and Water Pollution

Dr. Gulmi Chakraborty
Assiatant Professor

Air Pollution



➤ **Air pollution** corresponds to release into the atmosphere of various gases, finely divided solids, or finely dispersed liquid aerosols at rates that exceed the natural capacity of the environment to dissipate and dilute or absorb them.

➤ These substances may reach concentrations in the air that cause undesirable health, economic, or aesthetic effects.

➤ Airborne suspensions of extremely small solid or liquid particles called “particulates” (e.g., soot, dust, smokes, fumes, mists), especially those less than 10 micrometres (μm ; millionths of a metre) in size, are significant air pollutants because of their very harmful effects on human health. They are emitted by various industrial processes, coal- or oil-burning power plants, residential heating systems, and automobiles.

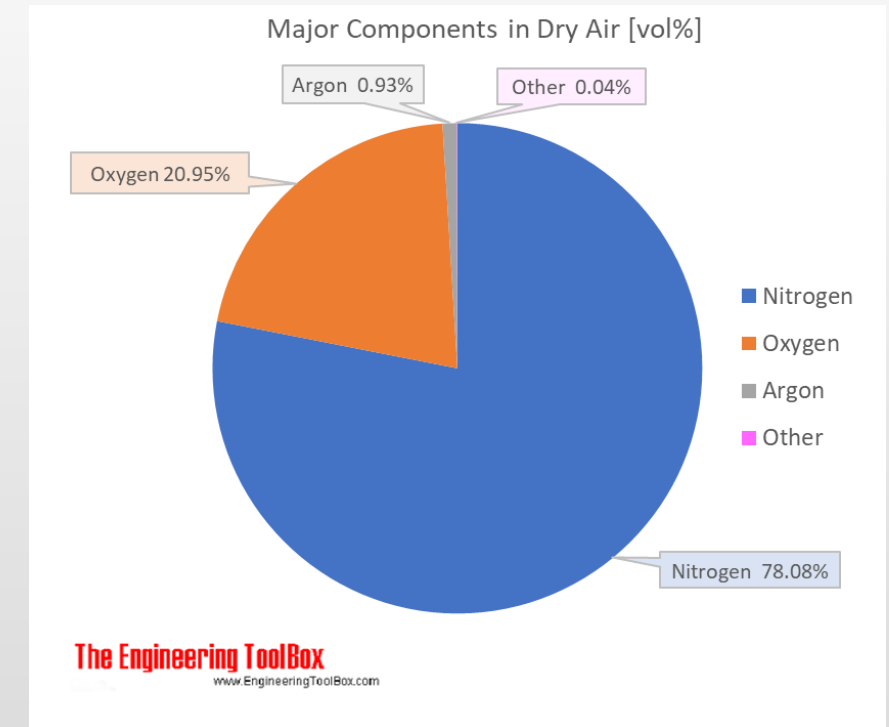


<https://education.nationalgeographic.org/resource/air-pollution/>

Major Air Pollutants (Criteria Pollutants)



- Clean, dry air consists primarily of nitrogen and oxygen—78 percent and 21 percent respectively, by volume.
- The remaining 1 percent is a mixture of other gases, mostly argon (0.9 percent), along with trace (very small) amounts of [carbon dioxide](#), [methane](#), [hydrogen](#), [helium](#), and more.
- Water vapour is also a normal, though quite variable, component of the atmosphere, normally ranging from 0.01 to 4 percent by volume; under very humid conditions the moisture content of air may be as high as 5 percent.
- There are six major air pollutants that have been designated by the U.S. [Environmental Protection Agency](#) (EPA) as “[criteria](#)” [pollutants](#)—*criteria* meaning that the concentrations of these pollutants in the atmosphere are useful as indicators of overall air quality.



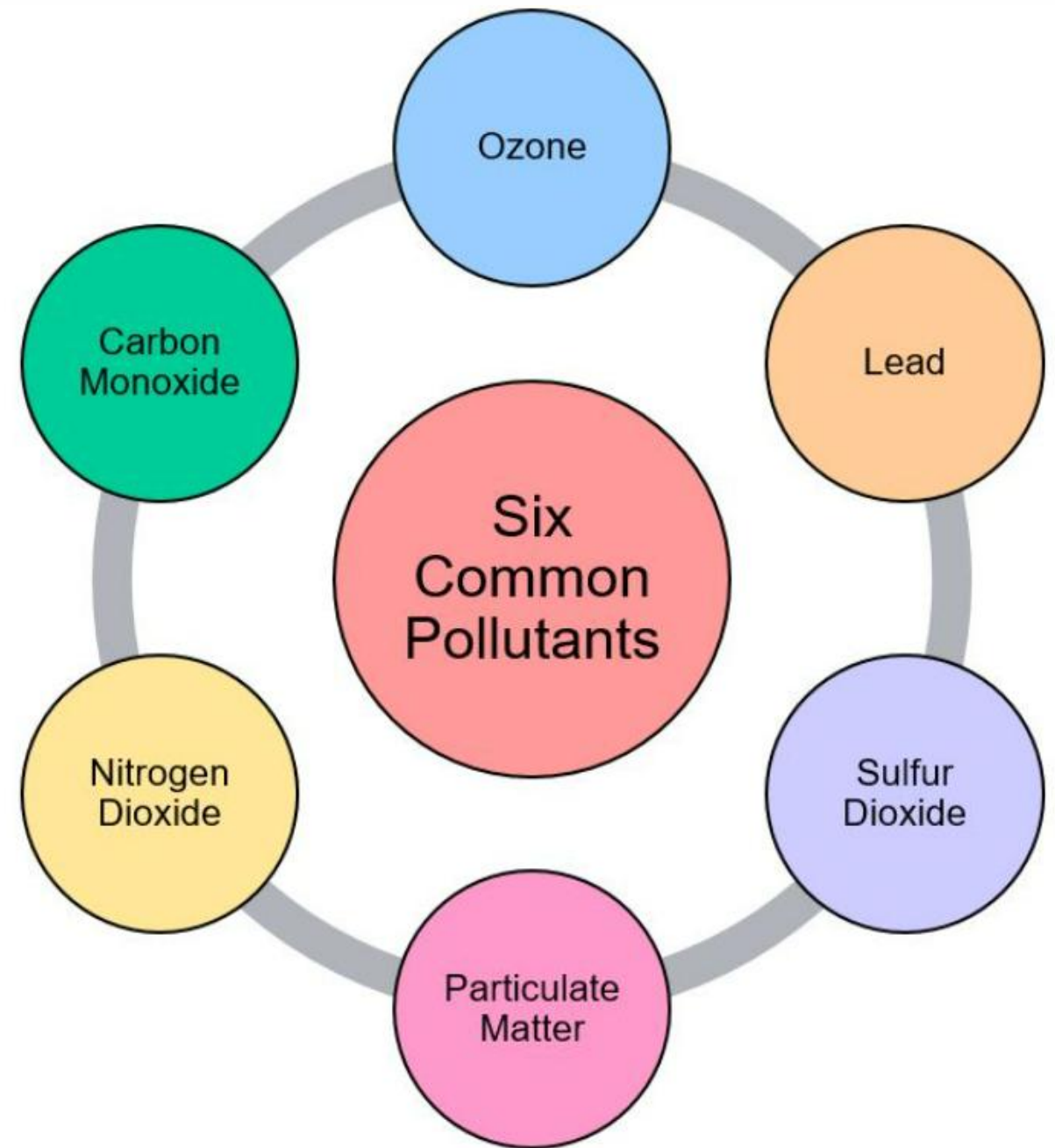
https://www.engineeringtoolbox.com/air-composition-d_212.html

Six Common Pollutants

The Clean Air Act requires EPA to set national ambient air quality standards (NAAQS) for specific pollutants to safeguard human health and the environment. These standards define the levels of air quality that EPA determines are necessary to protect against the adverse impacts of air pollution based on scientific evidence. EPA has established standards for six common air pollutants, which are referred to as “criteria” pollutants.

- **Carbon monoxide (CO)**
- **Lead (Pb)**
- **Nitrogen dioxide (NO₂)**
- **Ozone (O₃)**
- **Particulate matter (PM), and**
- **Sulfur dioxide (SO₂)**

<https://mydasdata.larc.nasa.gov/interactive-models/criteria-pollutants>

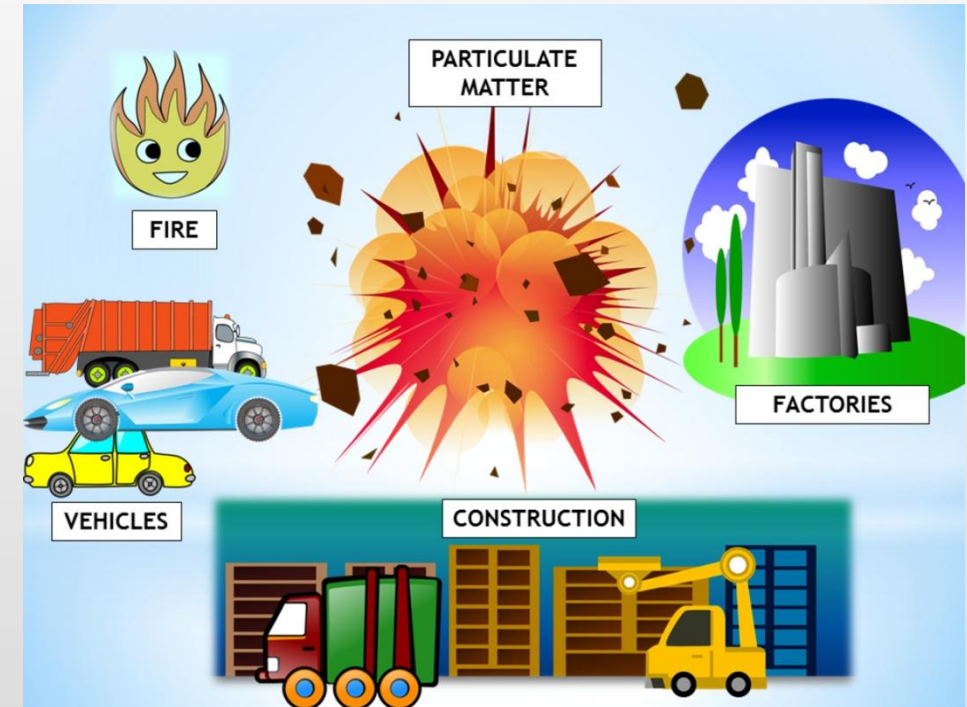


Sources of Air Pollution



➤The gaseous **criteria air pollutants** of primary concern in urban settings include sulfur dioxide, nitrogen dioxide, and carbon monoxide; these are emitted directly into the air from fossil fuels such as fuel oil, gasoline, and natural gas that are burned in power plants, automobiles, and other combustion sources.

➤Ozone (a key component of smog) is also a gaseous pollutant (**criteria Pollutant**); it forms in the atmosphere via complex chemical reactions occurring between nitrogen dioxide and various volatile organic compounds (e.g., gasoline vapours).

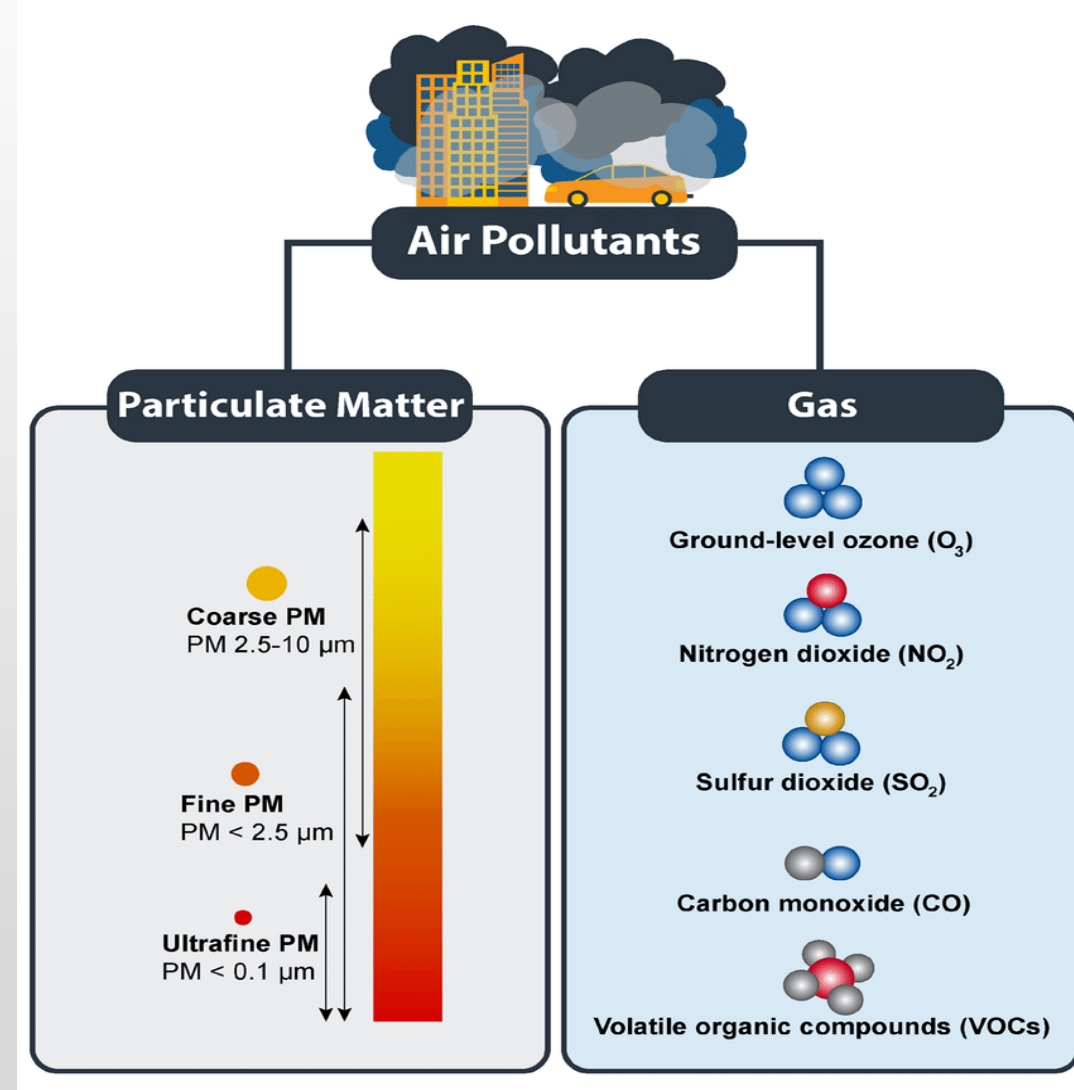


<https://soapboxie.com/social-issues/Sources-and-Effects-of-Major-Air-Pollutants-and-Their-Different-Pollution-Control-Systems>

Sources of Air Pollution



- Airborne suspensions of extremely small solid or liquid particles called “particulates” (**criteria pollutant**) (e.g., soot, dust, smokes, fumes, mists), especially those less than 10 micrometres (μm ; millionths of a metre) in size, are significant air pollutants because of their very harmful effects on human health.
- They are emitted by various industrial processes, coal- or oil-burning power plants, residential heating systems, and automobiles.



https://www.researchgate.net/figure/Classification-of-air-pollutant-according-to-some-physical-properties-Abbreviation-PM_fig1_339567258

- Lead (**criteria pollutant**) fumes (airborne particulates less than $0.5\text{ }\mu\text{m}$ in size) are particularly toxic and are an important pollutant of many diesel fuels.
- Solid particulates between roughly 1 and $100\text{ }\mu\text{m}$ in diameter are called dust particles, whereas airborne solids less than $1\text{ }\mu\text{m}$ in diameter are called fumes.
- The particulates of most concern with regard to their effects on human health are solids less than $10\text{ }\mu\text{m}$ in diameter, because they can be inhaled deep into the lungs and become trapped in the lower respiratory system.
- Certain particulates, such as asbestos fibres, are known carcinogens (cancer-causing agents), and many carbonaceous particulates—e.g., soot—are suspected of being carcinogenic.
- Major sources of particulate emissions include fossil-fuel power plants, manufacturing processes, fossil-fuel residential heating systems, and gasoline-powered vehicles.

Carbon monoxide (CO)



- Carbon monoxide is an odourless, invisible gas formed as a result of incomplete combustion. It is the most abundant of the criteria pollutants.
- Gasoline-powered highway vehicles are the primary source, although residential heating systems and certain industrial processes also emit significant amounts of this gas.
- Power plants emit relatively little carbon monoxide because they are carefully designed and operated to maximize combustion efficiency.
- Exposure to carbon monoxide can be acutely **harmful since it readily displaces oxygen in the bloodstream, leading to asphyxiation at high enough concentrations and exposure times.**

CARBON MONOXIDE THE SILENT KILLER



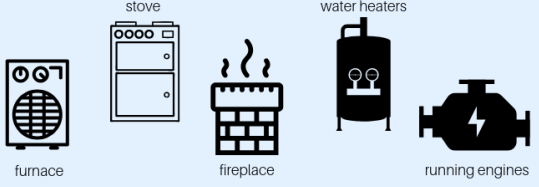
500
deaths
per year

check yourself
do you know the signs of CO poisoning?



nausea
vomiting
shortness of breath
headache

leading causes of carbon monoxide poisoning



furnace
stove
fireplace
water heaters
running engines



men are 2x more
likely to be affected
by CO poisoning
than women

https://www.cdc.gov/mmwr/preview/mmwrhtml/mm5650a1.htm?s_cid=mm5650a1_e#tab1
<https://www.cdc.gov/features/copoisoning/index.html>
<https://www.mayoclinic.org/diseases-conditions/carbon-monoxide/symptoms-causes/syc-20370642>



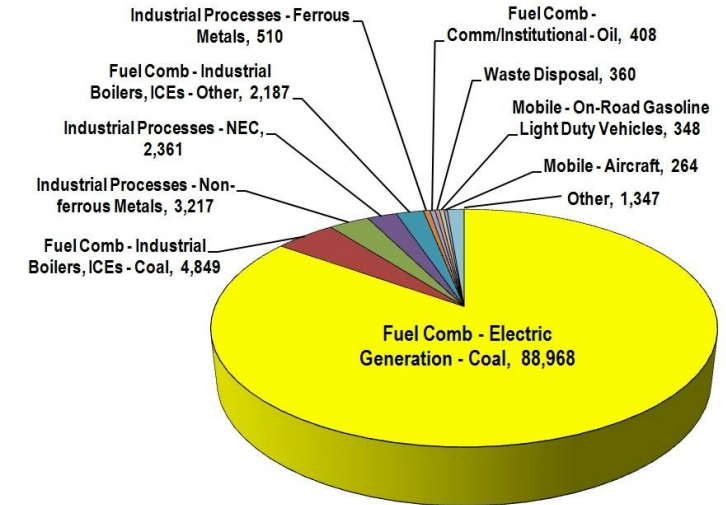
https://workplacelearningsystem.com/Safety-System-News_Blog/carbon-monoxide-safety.aspx

Sulfur Dioxide (SO₂)

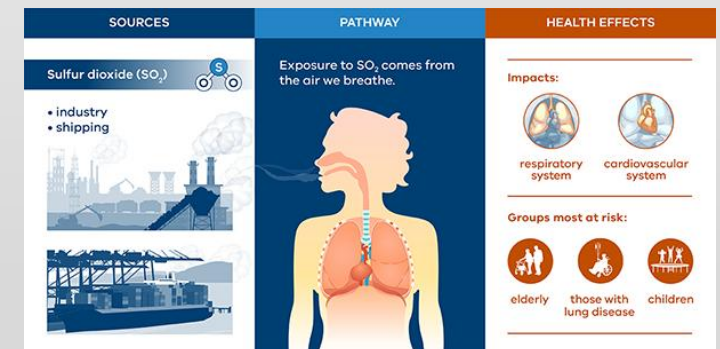


- A colourless gas with a sharp, choking odour, sulfur dioxide is formed during the combustion of coal or oil that contains sulfur as an impurity.
- Most sulfur dioxide emissions come from **power-generating plants**; very little comes from mobile sources.
- This pungent gas can cause **eye and throat** irritation and harm lung tissue when inhaled.
- Sulfur dioxide also reacts with oxygen and water vapour in the air, forming a mist of sulfuric acid that reaches the ground as a component of acid rain.
- Acid rain is believed to have harmed or destroyed fish and plant life.
- It also causes corrosion of metals and deterioration of the exposed surfaces of buildings and public monuments.

Top Sources in 7 County Area for SO₂ Emissions (104816 tons/year)



<https://caphedetroit.sph.umich.edu/project/sulfur-dioxide/>



<https://www.epa.vic.gov.au/for-community/environmental-information/air-quality/sulfur-dioxide-in-the-air>

Nitrogen Dioxide (NO₂)



- Of the several forms of nitrogen oxides, nitrogen dioxide—a pungent, irritating gas—is of most concern. It is known to cause pulmonary edema, an accumulation of excessive fluid in the lungs.
- Nitrogen dioxide also reacts in the atmosphere to form nitric acid, contributing to the problem of acid rain.
- In addition, nitrogen dioxide plays a role in the formation of photochemical smog, a reddish brown haze that often is seen in many urban areas and that is created by sunlight-promoted reactions in the lower atmosphere.
- Nitrogen oxides are formed when combustion temperatures are high enough to cause molecular nitrogen in the air to react with oxygen.
- Stationary sources such as coal-burning power plants are major contributors of this pollutant, although gasoline engines and other mobile sources are also significant.

Fog and Smog



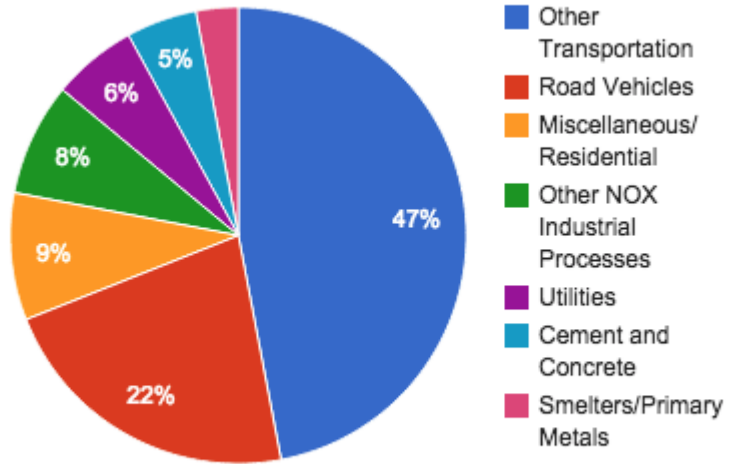
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Global University
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- Fog is due to water droplets in air.
- Smog is smoke + fog.

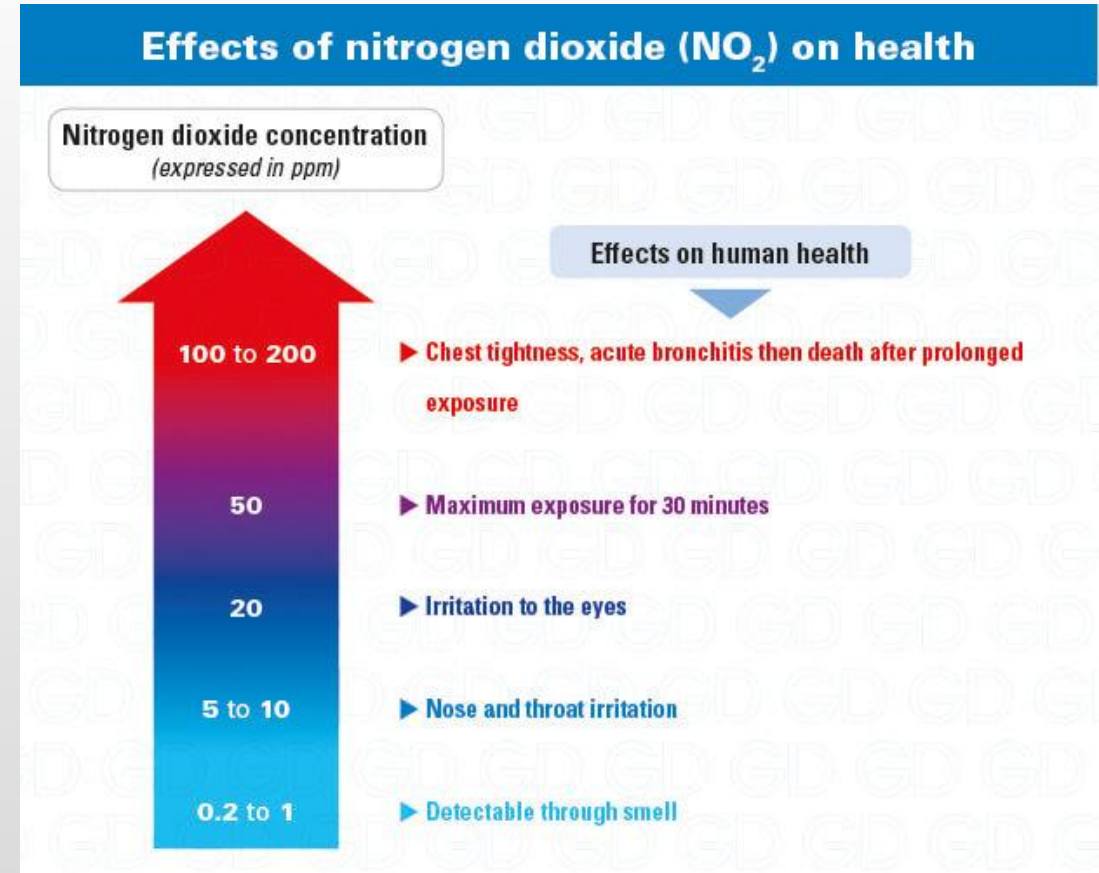


<https://www.pranaair.com/blog/difference-between-fog-and-smog/>

Nitrogen Dioxide (NO₂)

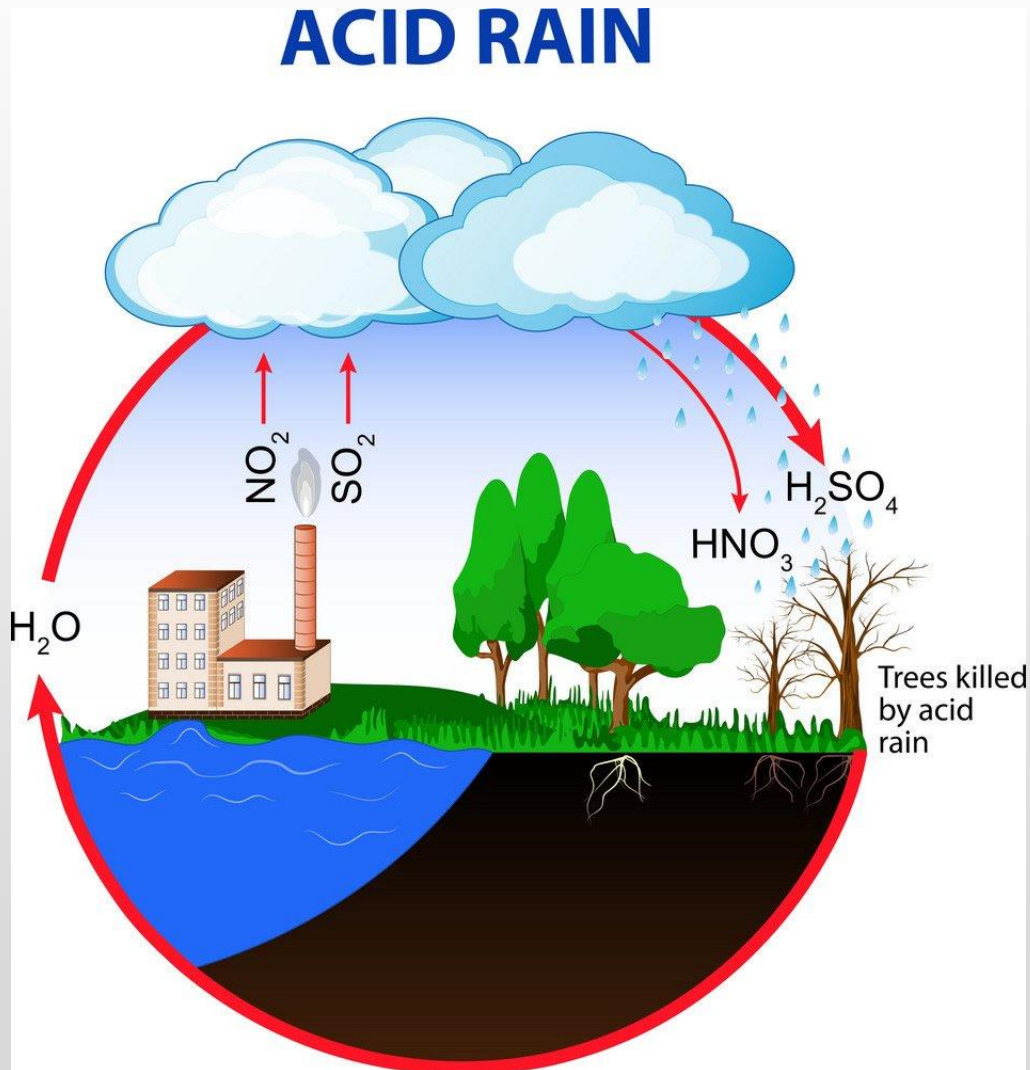


<https://cleanairhamilton.ca/nitrogen-oxide/>

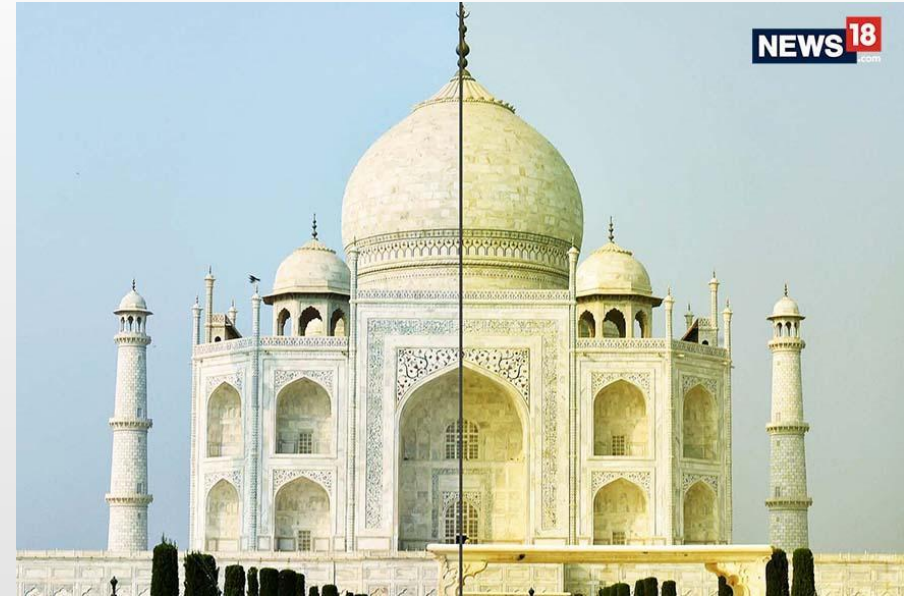


https://eportal.incometax.gov.in/iec/foreservices/#/foreturns-ay23/fy_itr1_2023/other_pages_itr1_2023/Return_Summary_ITR1

Acid rain



<https://www.internetgeography.net/topics/what-is-acid-rain/>



<https://www.thoughtco.com/acid-rain-definition-1434936>

Ozone (O₃)



- A key component of **photochemical smog**, ozone is formed by a complex reaction between nitrogen dioxide and hydrocarbons in the presence of sunlight.
- It is considered to be a criteria pollutant in the troposphere—the lowermost layer of the atmosphere—but not in the upper atmosphere, where it occurs naturally and serves to block harmful ultraviolet rays from the Sun.
- Because nitrogen dioxide and hydrocarbons are emitted in significant quantities by motor vehicles, photochemical smog is common in cities such as Los Angeles, where sunshine is ample and highway traffic is heavy.
- Certain geographic features, such as mountains that impede air movement, and weather conditions, such as temperature inversions in the troposphere, contribute to the trapping of air pollutants and the formation of photochemical smog.

- Inhaled lead particulates in the form of fumes and dusts are particularly harmful to children, in whom even slightly elevated levels of lead in the blood can cause learning disabilities, seizures, or even death (see lead poisoning).
- Sources of airborne lead particulates include oil refining, smelting, and other industrial activities.
- In the past, combustion of gasoline containing a lead-based antiknock additive called tetraethyl lead was a major source of lead particulates.
- In many countries there is now a complete ban on the use of lead in gasoline.
- In the United States, lead concentrations in outdoor air decreased more than 90 percent after the use of leaded gasoline was restricted in the mid-1970s and then completely banned in 1996.

- Hundreds of specific substances are considered hazardous when present in trace amounts in the air. These pollutants are called air toxics.
- Many of them cause genetic mutations or [cancer](#); some cause other types of health problems, such as adverse effects on brain tissue or fetal development.
- Although the total emissions and the number of sources of air toxics are small compared with those for criteria pollutants, these pollutants can pose an immediate health risk to exposed individuals and can cause other environmental problems.

- Most air toxics are [organic chemicals](#), [comprising](#) molecules that contain [carbon](#), [hydrogen](#), and other atoms. Many are [volatile organic compounds](#) (VOCs), organic [compounds](#) that readily evaporate.
- VOCs include pure [hydrocarbons](#), partially oxidized hydrocarbons, and organic compounds containing [chlorine](#), [sulfur](#), or [nitrogen](#).
- They are widely used as fuels (e.g., [propane](#) and [gasoline](#)), as [paint](#) thinners and solvents, and in the production of [plastics](#).
- In addition to contributing to air toxicity and urban smog, some VOC emissions act as [greenhouse gases](#) and, in so doing, contribute to [global warming](#).
- Some other air toxics are metals or compounds of metals—for example, [mercury](#), [arsenic](#), and [cadmium](#).

- Air toxics may be released in sudden and catastrophic accidents rather than steadily and gradually from many sources. For example, in the [Bhopal disaster](#) of 1984, an accidental release of methyl isocyanate at a [Union Carbide pesticide](#) factory in Bhopal, [Madhya Pradesh](#) state, [India](#), immediately killed at least 3,000 people, eventually caused the deaths of an estimated 15,000 to 25,000 people over the following quarter-century, and injured hundreds of thousands more.
- The risk of accidental release of very hazardous substances into the air is generally higher for people living in industrialized urban areas. Hundreds of such incidents occur each year, though none has been as severe as the Bhopal event.

- Other than in cases of occupational exposure or accidental release, health threats from air toxics are greatest for people who live near large industrial facilities or in congested and polluted urban areas.
- Most major sources of air toxics are so-called point sources—that is, they have a specific location. Point sources include chemical plants, steel mills, oil refineries, and municipal waste incinerators.
- Hazardous air pollutants may be released when equipment leaks or when material is transferred, or they may be [emitted](#) from smokestacks.
- Municipal waste incinerators, for example, can emit hazardous levels of [dioxins](#), [formaldehyde](#), and other organic substances, as well as metals such as arsenic, beryllium, lead, and mercury.
- Nevertheless, proper combustion along with appropriate [air pollution control](#) devices can reduce emissions of these substances to acceptable levels.

- Hazardous air pollutants also come from “area” sources, which are many smaller sources that release pollutants into the outdoor air in a defined area.
- Such sources include commercial [dry-cleaning](#) facilities, gasoline stations, small metal-plating operations, and woodstoves.
- Emission of air toxics from area sources are also regulated under some circumstances.
- Small area sources account for about 25 percent of all emissions of air toxics. Major point sources account for another 20 percent.
- The rest—more than half of hazardous air-pollutant emissions—come from motor vehicles. For example, [benzene](#), a component of gasoline, is released as unburned fuel or as fuel vapours, and [formaldehyde](#) is one of the [by-products](#) of incomplete combustion.
- Newer cars, however, have emission control devices that significantly reduce the release of air toxics.

- Global warming is recognized by almost all atmospheric scientists as a significant environmental problem caused by an increase in levels of certain trace gases in the atmosphere since the beginning of the Industrial Revolution in the mid-18th century.
- These gases, collectively called greenhouse gases, include carbon dioxide, organic chemicals called chlorofluorocarbons (CFCs), methane, nitrous oxide, ozone, and many others. Carbon dioxide, although not the most potent of the greenhouse gases, is the most important because of the huge volumes emitted into the air by combustion of fossil fuels (e.g., gasoline, oil, coal).

Air Pollution Control



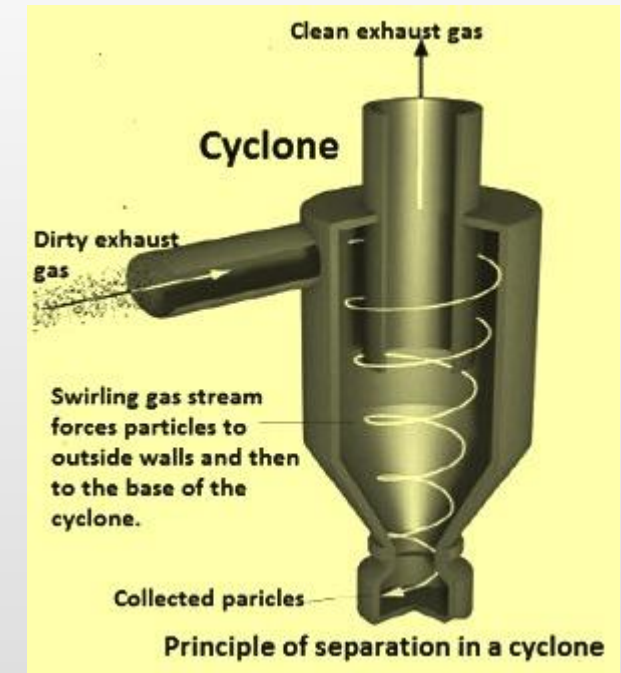
- The techniques employed **to reduce or eliminate the emission into the atmosphere** of substances that can harm the environment or human health.
- The control of air pollution is one of the principal areas of pollution control, along with wastewater treatment, solid-waste management, and hazardous-waste management.
- The best way to protect air quality is **to reduce the emission** of pollutants by changing to cleaner fuels and processes.
- Pollutants not eliminated in this way must **be collected or trapped by appropriate air-cleaning devices** as they are generated and before they can escape into the atmosphere.



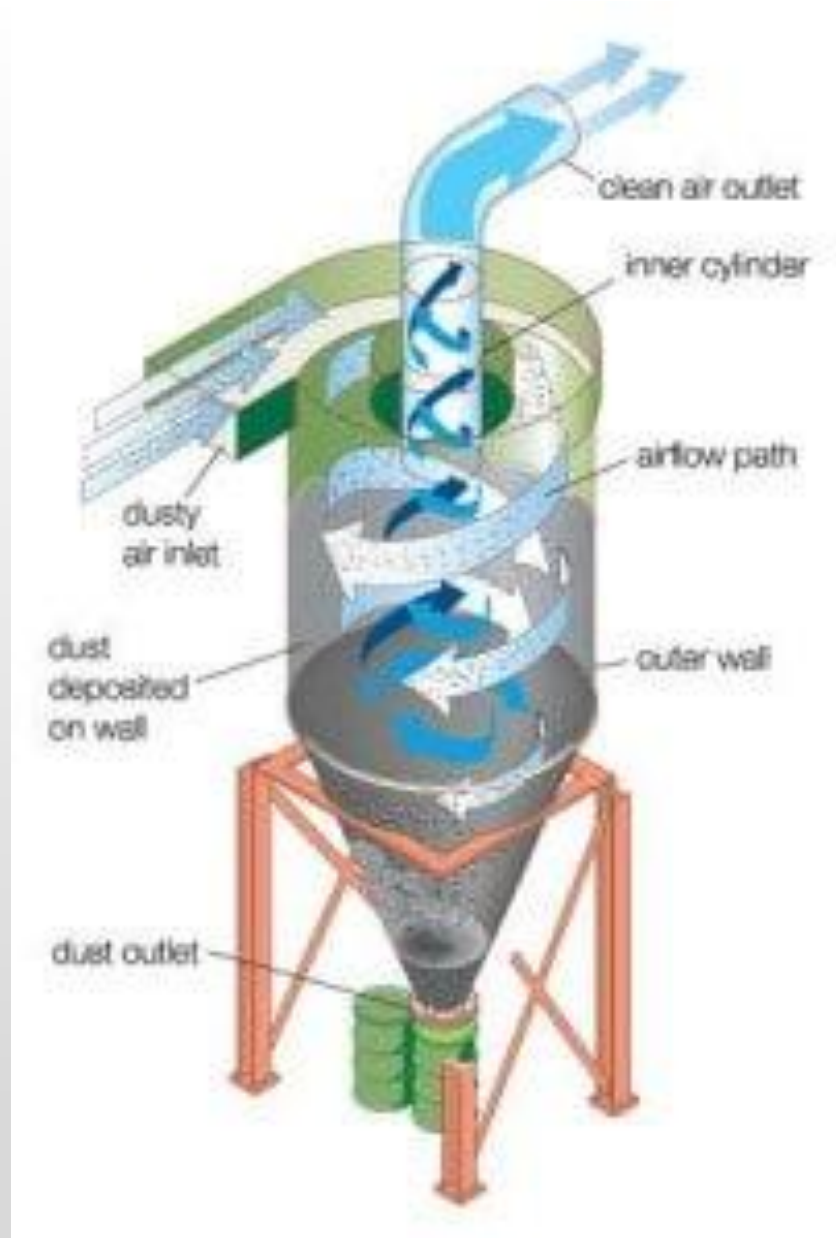
<https://www.eletimes.com/uk-energy-giants-pivot-towards-cleaner-fuels>

- Common types of equipment for collecting fine particulates include cyclones, scrubbers, electrostatic precipitators, and baghouse filters.
- Once collected, particulates adhere to each other, forming agglomerates that can readily be removed from the equipment and disposed of, usually in a landfill.

- A cyclone removes particulates by causing the dirty airstream to flow in a spiral path inside a cylindrical chamber.
- Dirty air enters the chamber from a tangential direction at the outer wall of the device, forming a vortex as it swirls within the chamber.
- The larger particulates, because of their greater inertia, move outward and are forced against the chamber wall.
- Slowed by friction with the wall surface, they then slide down the wall into a conical dust hopper at the bottom of the cyclone.
- The cleaned air swirls upward in a narrower spiral through an inner cylinder and emerges from an outlet at the top.



<https://www.ispatguru.com/air-pollution-control-devices/>



- A cyclone separator is a separation device that uses the principle of inertia to remove particulate matter from flue gases.
- In these separators, dirty flue gas enters the chamber from a tangential direction at the outer wall of the device, forming a vortex as it swirls within the chamber.
- The larger particulates, because of their greater inertia, move outward and are forced against the chamber wall and drop down.
- This separates the particulate matter from the flue gas, leaving cleaned flue gas. Cyclone separators are also used as pre treatment to other air pollution control devices to increase their efficiency.

Uses of Cyclone Separators

Precleaner with Fabric
Filters/ESP/ Wet Scrubber

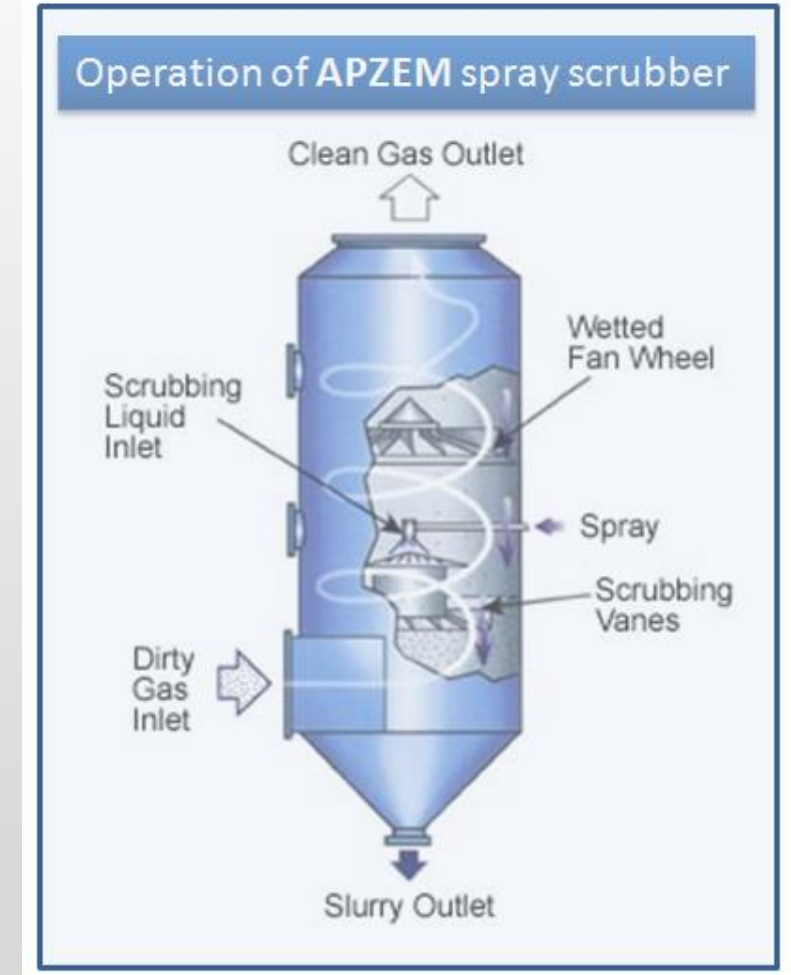
Recovery and Recycling of
Process Material

Spray drying in Food and
Chemical Industries

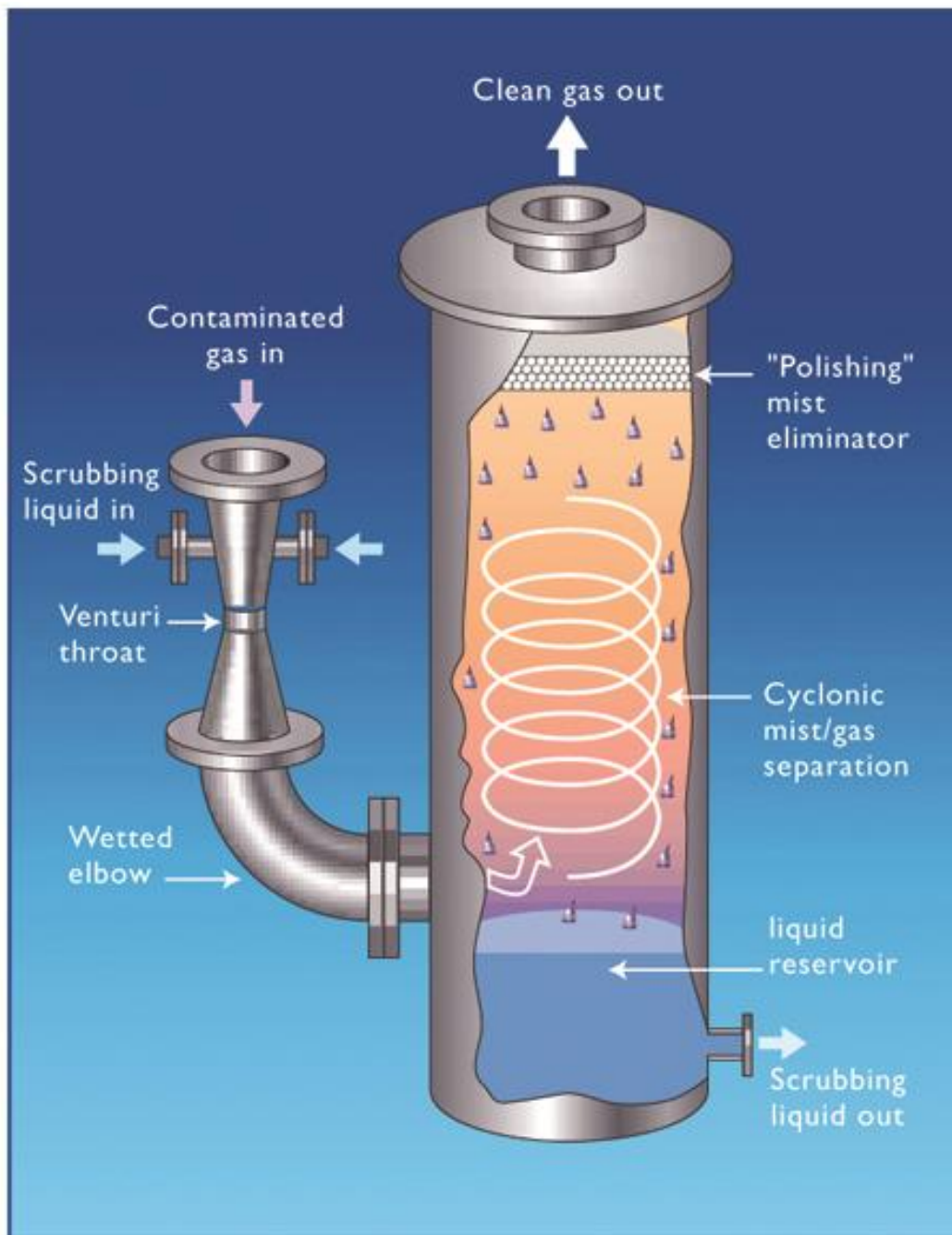
First stage control of PM
emissions from kilns and
furnaces

- Accumulated particulate dust is periodically removed from the hopper for disposal.
- Cyclones are best at removing relatively coarse particulates.
- They can routinely achieve efficiencies of 90 percent for particles larger than about 20 micrometres (μm ; 20 millionths of a metre).
- By themselves, however, cyclones are not sufficient to meet stringent air quality standards.
- They are typically used as pre-cleaners and are followed by more efficient air-cleaning equipment such as electrostatic precipitators and baghouses (described below).

- Devices called wet scrubbers trap suspended particles by direct contact with a spray of water or other liquid.
- In effect, a scrubber washes the particulates out of the dirty airstream as they collide with and are entrained by the countless tiny droplets in the spray.
- Spray-tower scrubbers can remove 90 percent of particulates larger than about 8 μm .



<https://www.apzem.com/india/products/spray-scrubber.html>



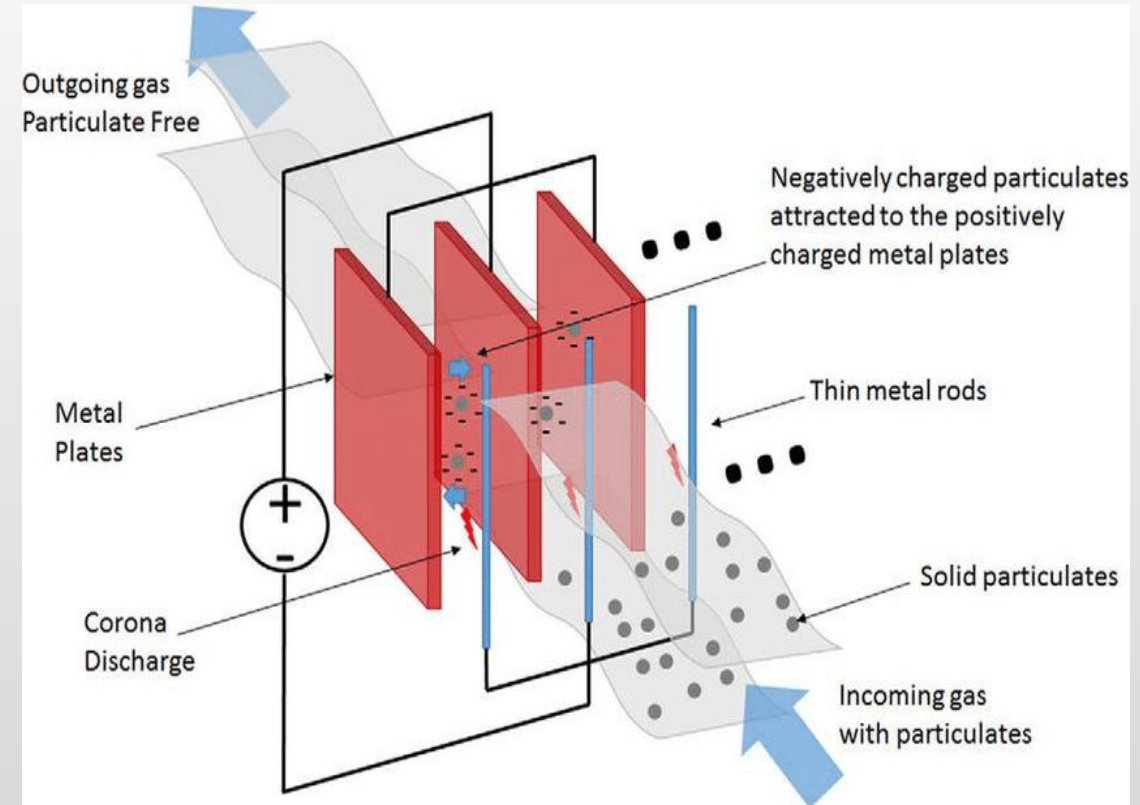


- Electrostatic precipitation is a commonly used method for removing fine particulates from airstreams. In an electrostatic precipitator, particles suspended in the airstream are given an electric charge as they enter the unit and are then removed by the influence of an electric field.
- The precipitation unit comprises baffles for distributing airflow, discharge and collection electrodes, a dust clean-out system, and collection hoppers.
- A high voltage of direct current (DC), as much as 100,000 volts, is applied to the discharge electrodes to charge the particles, which then are attracted to oppositely charged collection electrodes, on which they become trapped.

Electrostatic precipitators



- In a typical unit the collection electrodes comprise a group of large rectangular metal plates suspended vertically and parallel to each other inside a boxlike structure.
- There are often hundreds of plates having a combined surface area of tens of thousands of square metres.
- Rows of discharge electrode wires hang between the collection plates.
- The wires are given a negative electric charge, whereas the plates are grounded and thus become positively charged.



- Particles that stick to the collection plates are removed periodically when the plates are shaken, or “rapped.” Rapping is a mechanical technique for separating the trapped particles from the plates, which typically become covered with a 6-mm (0.2-inch) layer of dust.
- The dislodged particles are collected in a hopper at the bottom of the unit and removed for disposal.
- An electrostatic precipitator can remove particulates as small as $1\text{ }\mu\text{m}$ with an efficiency exceeding 99 percent.
- The effectiveness of electrostatic precipitators in removing fly ash from the combustion gases of fossil-fuel furnaces accounts for their high frequency of use at power stations.



Baghouse Filters



- One of the most efficient devices for removing suspended particulates is an assembly of fabric-filter bags, commonly called a baghouse.
- A typical baghouse comprises an array of long, narrow bags—each about 25 cm (10 inches) in diameter—that are suspended upside down in a large enclosure.
- Dust-laden air is blown upward through the bottom of the enclosure by fans.
- Particulates are trapped inside the filter bags, while the clean air passes through the fabric and exits at the top of the baghouse.
- A fabric-filter dust collector can remove very nearly 100 percent of particles as small as $1\ \mu\text{m}$ and a significant fraction of particles as small as $0.01\ \mu\text{m}$.



<https://www.nachi.org/baghouse-inspection.htm>



1,500 × 1,127





This is an excerpt from our Baghouse Basics training module.
Our EHS videos can be found at www.convergenceeltraining.com

Baghouse Filters

- Fabric filters, however, offer relatively high resistance to airflow, which leads to substantial energy usage for the fan system.
- In addition, in order to prolong the useful life of the filter fabric, the air to be cleaned must be cooled (usually below 300 °C [570 °F]) before it is passed through the unit; cooling coils needed for this purpose add to the energy usage. (Certain filter fabrics—e.g., those made of ceramic or mineral materials—can operate at higher temperatures.)
- Several compartments of filter bags are often used at a single baghouse installation. This arrangement allows individual compartments to be cleaned while others remain in service. The bags are cleaned by removing the excess layer of surface dust.

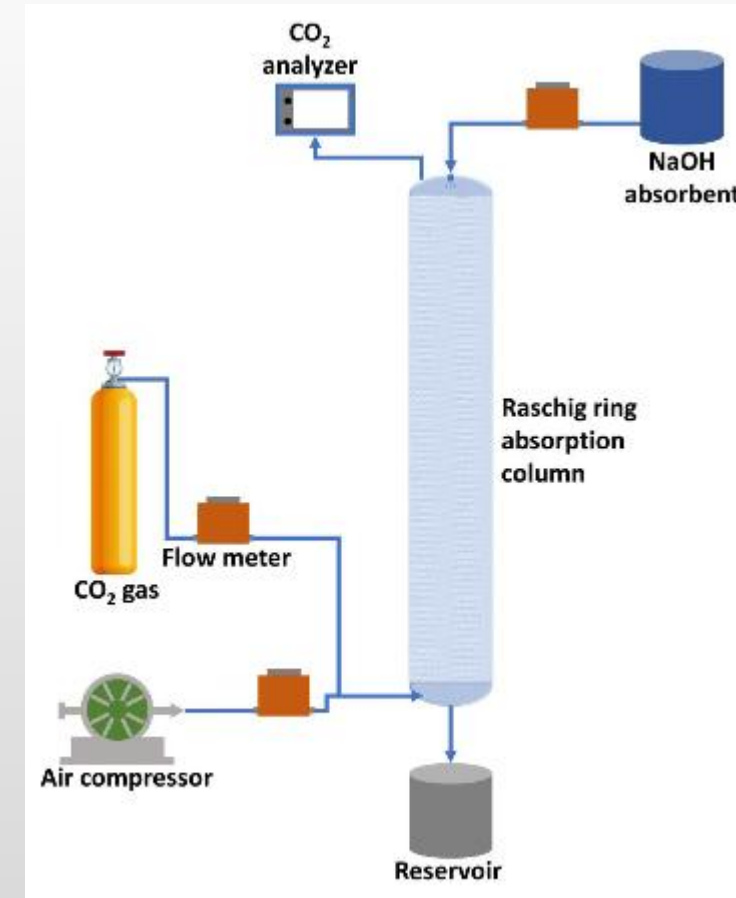
- This is done in several different ways: by mechanically shaking them; by temporarily reversing the flow of air and causing them to collapse; or by sending a short burst of air down through the bag, causing it to briefly expand.
- After the dust is removed from the filters, it falls into a hopper below and can be collected for disposal or further use.
- Care must be taken not to remove too much of the built-up surface dust, or “dust cake,” when cleaning the filters.
- In most filter types the filter itself is only a substrate that allows for the formation of a layer of dust cake, which then captures the majority of the particulates.
- Filters with an applied membrane coating such as polytetrafluoroethylene (Teflon) do not require the use of dust cake to operate at their highest efficiency.

- Gaseous criteria pollutants, as well as volatile organic compounds (VOCs) and other gaseous air toxics, are controlled by means of three basic techniques: absorption, adsorption, and incineration (or combustion).
- These techniques can be employed singly or in combination. T
- They are effective against the major greenhouse gases as well.
- In addition, a fourth technique, known as carbon sequestration, is in development as a means of controlling carbon dioxide levels.

Absorption



- In the context of air pollution control, absorption involves the transfer of a gaseous pollutant from the air into a contacting liquid, such as water.
- The liquid must be able either to serve as a solvent for the pollutant or to capture it by means of a chemical reaction.



<https://www.mdpi.com/2411-5134/8/3/70>

Wet Scrubbers and Packed Scrubbers



- Wet scrubbers similar to those described above for controlling suspended particulates may be used for gas absorption.
- A packed scrubber is the countercurrent tower. After entering the bottom of the tower, the polluted airstream flows upward through a wetted column of light, chemically inactive packing material.
- The liquid absorbent flows downward and is uniformly spread throughout the column packing, thereby increasing the total area of contact between gas and liquid.
- Thermoplastic materials are most widely used as packing for countercurrent scrubber towers. These devices usually have gas-removal efficiencies of 90–95 percent.



<https://m.indiamart.com/proddetail/packed-bed-tower-scrubbers-22846720497.html>

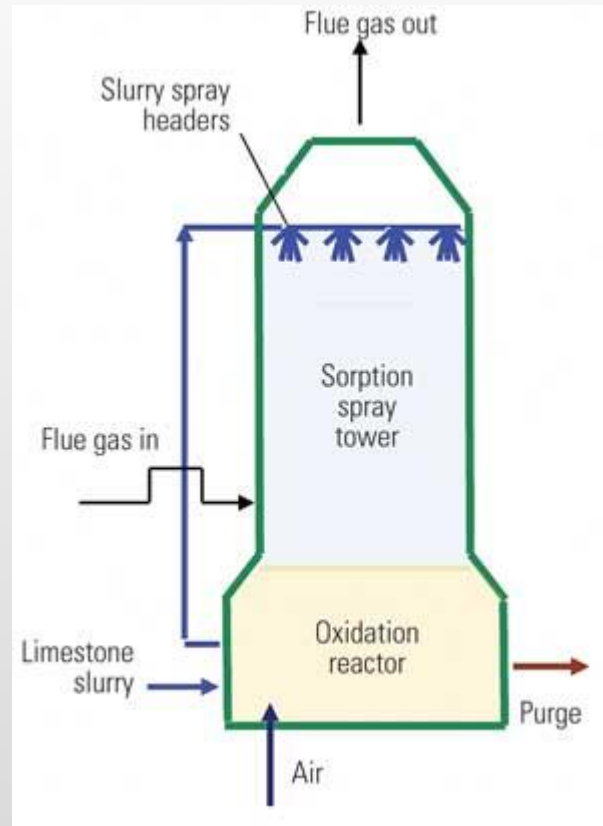
Wet Scrubbers and Packed Scrubbers



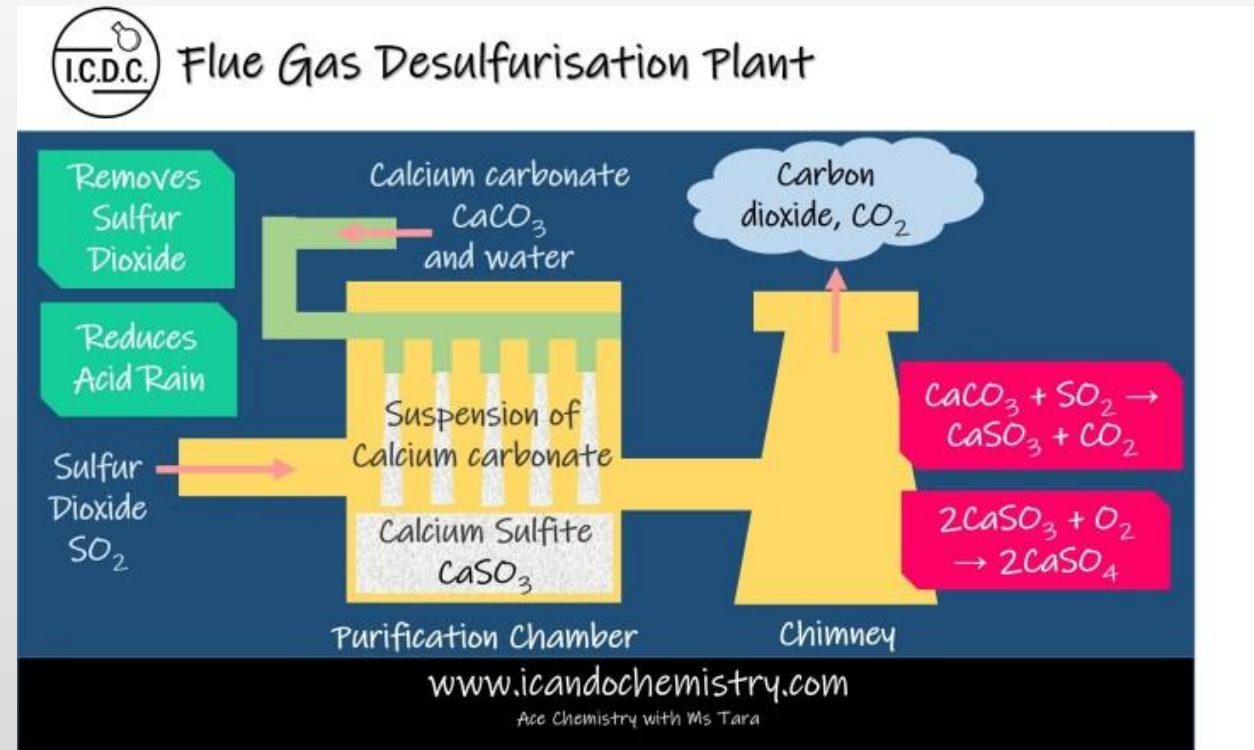
➤ In general, scrubbers are used at fertilizer production facilities (to remove ammonia from the airstream), at glass production plants (to remove hydrogen fluoride), at chemical plants (to remove water-soluble solvents such as acetone and methyl alcohol), and at rendering plants (to control odours).

- Sulfur dioxide in flue gas from fossil-fuel power plants can be controlled by means of an absorption process called flue gas desulfurization (FGD).
- FGD systems may involve wet scrubbing or dry scrubbing.
- In wet FGD systems, flue gases are brought in contact with an absorbent, which can be either a liquid or a slurry of solid material.
- The sulfur dioxide dissolves in or reacts with the absorbent and becomes trapped in it.
- In dry FGD systems, the absorbent is dry pulverized lime or limestone; once absorption occurs, the solid particles are removed by means of baghouse filters (described above).
- Dry FGD systems, compared with wet systems, offer cost and energy savings and easier operation, but they require higher chemical consumption and are limited to flue gases derived from the combustion of low-sulfur coal.

Flue Gas Desulfurization



<https://www.powermag.com/flue-gas-desulfurization-wastewater-treatment-primer/>



- In the limestone scrubbing process, sulfur dioxide reacts with limestone (calcium carbonate) particles in the slurry, forming calcium sulfite and carbon dioxide.
- In the lime scrubbing process, sulfur dioxide reacts with slaked lime (calcium hydroxide), forming calcium sulfite and water.
- Depending on sulfur dioxide concentrations and oxidation conditions, the calcium sulfite can continue to react with water, forming calcium sulfate (gypsum).
- Neither calcium sulfite nor calcium sulfate is very soluble in water, and both can be precipitated out as a slurry by gravity settling.
- The thick slurry, called FGD sludge, creates a significant disposal problem. Flue gas desulfurization helps to reduce ambient sulfur dioxide levels and mitigate the problem of acid rain. Nevertheless, in addition to its expense (which is passed on directly to the consumer as higher rates for electricity), millions of tons of FGD sludge are generated each year.

- Gas adsorption, as contrasted with absorption, is a surface phenomenon. The gas molecules are sorbed— attracted to and held—on the surface of a solid.
- Gas adsorption methods are used for odour control at various types of chemical-manufacturing and food-processing facilities, in the recovery of a number of volatile solvents (e.g., benzene), and in the control of VOCs at industrial facilities.
- Activated carbon (heated charcoal) is one of the most common adsorbent materials. It is very porous and has an extremely high ratio of surface area to volume.
- Activated carbon is particularly useful as an adsorbent for cleaning airstreams that contain VOCs and for solvent recovery and odour control.
- A properly designed carbon adsorption unit can remove gas with an efficiency exceeding 95 percent.

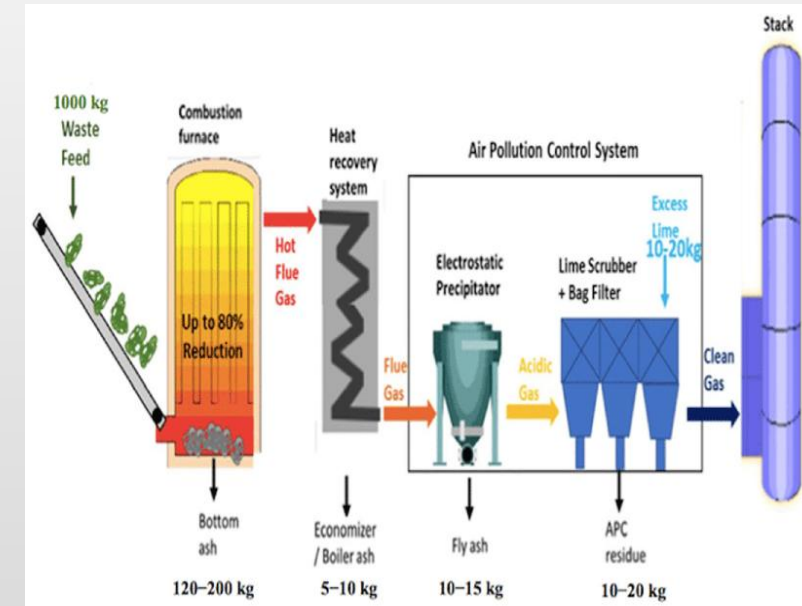


<https://youtu.be/BAfoMXruUHA>

- The process called incineration or combustion—chemically, rapid oxidation—can be used to convert VOCs and other gaseous hydrocarbon pollutants to carbon dioxide and water.
- Incineration of VOCs and hydrocarbon fumes usually is accomplished in a special incinerator called an afterburner.
- To achieve complete combustion, the afterburner must provide the proper amount of turbulence and burning time, and it must maintain a sufficiently high temperature.
- Sufficient turbulence, or mixing, is a key factor in combustion because it reduces the required burning time and temperature.
- A process called direct flame incineration can be used when the waste gas is itself a combustible mixture and does not need the addition of air or fuel.

Incineration

- An afterburner typically is made of a steel shell lined with refractory material such as firebrick. The refractory lining protects the shell and serves as a thermal insulator.
- Certain substances, such as platinum, can act in a manner that assists the combustion reaction.
- These substances, called catalysts, allow complete oxidation of the combustible gases at relatively low temperatures.
- Afterburners are used to control odours, destroy toxic compounds, or reduce the amount of photochemically reactive substances released into the air.
- They are employed at a variety of industrial facilities where VOC vapours are emitted from combustion processes or solvent evaporation (e.g., petroleum refineries, paint-drying facilities, and paper mills).

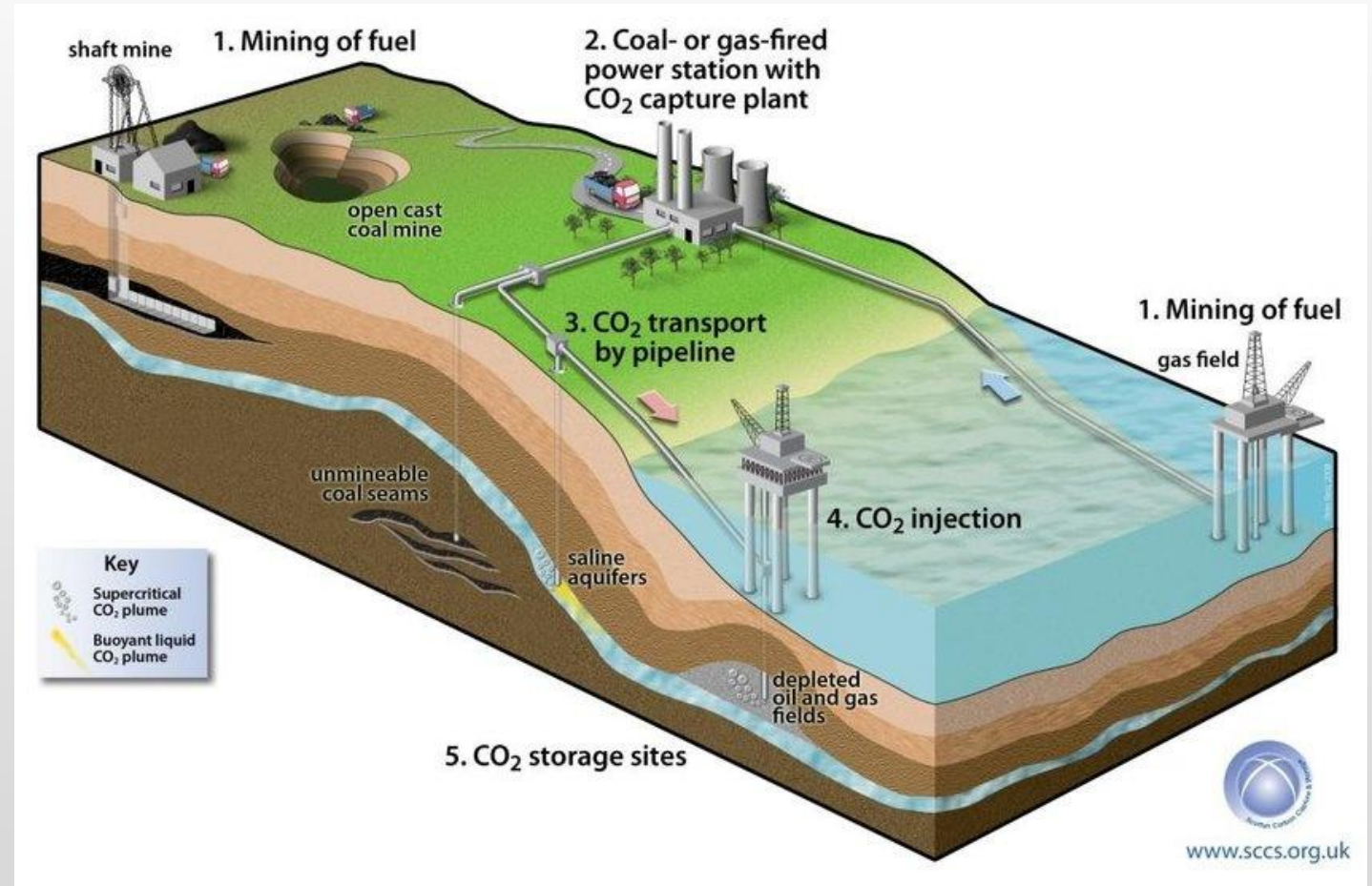


- The best way to reduce the levels of carbon dioxide in the air is to use energy more efficiently and to reduce the combustion of fossil fuels by using alternative energy sources (e.g., nuclear, wind, tidal, and solar power).
- In addition, carbon sequestration can be used to serve the purpose. Carbon sequestration involves the long-term storage of carbon dioxide underground, as well as on the surface of Earth in forests and oceans.
- Carbon sequestration in forests and oceans relies on natural processes such as forest growth. However, the clearing of forests for agricultural and other purposes (and also the pollution of oceans) diminishes natural carbon sequestration.

Carbon Sequestration



➤ Storing carbon dioxide underground—a technology under development that is also called geosequestration or carbon capture and storage—would involve pumping the gas directly into underground geologic “reservoir” layers. This would require the separation of carbon dioxide from power plant flue gases (or some other source)—a costly process.



https://www.researchgate.net/figure/Schematic-Diagram-of-the-Geosequestration-Process-Methane-gas-produced-from-offshore_fig11_327750644

Key Program goals include:

- Determining the CO₂ storage resource potential of on and offshore oil, gas, and saline bearing formations
- Improving carbon storage efficiency and security by advancing new and early-stage monitoring tools and models
- Improving capabilities to evaluate and manage environmental risks and uncertainty through integrated risk-based strategic monitoring and mitigation protocols
- Disseminating findings and lessons learned to the broader CCS community and key stakeholders

The program is comprised of three primary components: (Click on each for more information)

[Storage Infrastructure](#),
[Core Storage Research and Development](#), and
[Strategic Program Support](#).

Carbon capture technology and how it works

Technology that captures carbon dioxide from our atmosphere has existed for decades and is now being considered a key method for fighting climate change. So what does this technology look like and how does it work in practice?

Carbon capture and storage (CCS) technology is a form of [carbon sequestration](#) that's set to play a central role in helping us reach [net zero](#) by 2050.

Existing strategies to tackle climate change focus mainly on eliminating the carbon emissions from processes such as power generation or transport; but CCS looks at how carbon dioxide (CO₂) can be captured directly from the atmosphere, or at point of emission, and stored safely within the natural environment.

How does carbon capture technology work?

CCS takes two basic forms:

Biological carbon capture and storage: when the natural environment – such as forests and oceans – sequesters CO₂ from the atmosphere.

Artificial / Geological carbon capture and storage: when CO₂ as an emission is extracted from human-made processes and is stored in vast underground facilities.

Biological CCS happens on a much larger scale than geological CCS, but the technology to stimulate both has traditionally been viewed as expensive and unpractical at scale. This is changing, however, as investment and research into carbon capturing technologies takes off.

Types of carbon capture technology

1. Carbon sinks

Natural forms of CCS are called 'carbon sinks' and they are vast spaces where the natural habitats capture CO₂ from the atmosphere – these include forests, oceans, grasslands and wetlands.

Scientists, as well as environmental and conservation experts, recognise that the preservation and cultivation of carbon sinks could increase the amount of carbon taken from our atmosphere in the shortest space of time.

Grasslands and wetlands in particular have a much quicker turnaround for carbon storage, with coastal wetlands storing more carbon per hectare than other habitats like forests.¹

Where woodland is used, experts believe certain types of tree - such as birch or willow – are optimal for land-based carbon capture as they absorb more CO₂ comparatively than other tree species.

2. Saline aquifers

Deep saline aquifers are underground geological formations; vast expanses of porous, sedimentary rock, which are filled with salt water. CO₂ can be injected into these and stored permanently – in fact, saline aquifers have the largest identified storage potential among all other forms of engineered CCS.

The 'Endurance' aquifer, located in the North Sea off the coast of the UK, is one such formation, which sits approximately 1 mile (1.6km) below the sea bed. Roughly the size of Manhattan Island and the height of The Shard or the Empire State Building, its porous composition allows for carbon dioxide to be injected into it and stored safely for potentially thousands of years.

In the US, multiple large-scale saline aquifers are now being used for CCS purposes, such as the [Citronelle Project](#) in Alabama. During its three-year trial period, it was successful in storing more than 150,000 tonnes of CO₂ per year, which was captured from a nearby pilot facility.

3. Giant air filters

Carbon capture technologies are still being developed globally, with individual countries creating strategies that respond to their own net zero goals. For example, in China companies have developed experimental commercial air filters – huge towers that clean air of pollutants on a huge scale. These giant air towers purify air by drawing it into glass rooms, which are heated using solar power creating a greenhouse effect. This hot air up is pushed up the tower through a series of filters, before being released back into the atmosphere as clean air.

One such giant air-purifier tower in Xian has reportedly been cleaning more than 353 million cubic feet of air each day, dramatically improving local air quality. Manufacturers believe they are close to developing even larger towers, where just one could clean enough air on a daily basis for a small city.

4. Ionic liquids: a capturing carbon technology of the future?

The most recent advancements in CCS technology includes new types of liquids, which are highly effective at absorbing CO₂. Two dimensional ‘ionic’ liquids have a molecular structure that allow for higher rates of CO₂ to be absorbed. Scientists believe ‘editing’ liquids can offer more precise control in the chemical engineering process and are considered environmentally friendly.

Helping to 'grow' biological CCS

- Planting trees or developing wetland areas are two of the simplest ways of stimulating CCS on huge scales, but there are often issues around land ownership and space.
- Governments, landowners and local communities all have a role to play in identifying opportunities for projects and collaboration.
- In the UK, financial incentives in the form of Environmental Land Management schemes are offered to landowners and land managers who commit to managing their land in an environmentally sustainable way.
- In the US, land reclamation for environmental purposes is similarly incentivised by the Department of Agricultural Farm Service Agency (FSA).
- Farmers can receive rental payments if they no longer develop on land or use it for agricultural purposes; allowing grasslands, uplands, wetlands and forests to flourish instead.

Helping to 'grow' biological CCS

- Businesses in the private sector are increasingly 'offsetting' the emissions they produce by funding carbon capture schemes.
- This is also known becoming carbon neutral through the purchase of 'carbon credits'.
- Microsoft, for example, has made its initial offset purchase through a \$1 billion investment in Climeworks' Orca CCS facility, located in Reykjavik.
- Climeworks opened the first commercial CCS facility in Switzerland in 2017, before opening the larger Orca site in 2021.
- It's estimated that the Icelandic plant can capture up to 4,000 metric tonnes from the atmosphere annually.

- The [Energy Conservation \(Amendment\) Bill, 2022](#), which was introduced and passed in the Lok Sabha on August 8, incorporated the aim of developing the domestic carbon market.
- While the bill focuses on the renewable energy industry, it can [indirectly](#) benefit the country's farmers as well.
- Approximately [55 percent](#) of India's population is employed in agriculture. Given its dependence on natural resources such as soil and water, agriculture is significantly impacted by the climate crisis and its accompanying effects including land degradation, market volatility in prices, and rising input costs. This affects the sustainability of agricultural production and the livelihoods of those dependent on it.

Improving soil health is critical

- The intensive application of fertilisers and pesticides since the Green Revolution has [depleted](#) soil carbon levels and consequently degraded the soil.
- Various [estimates](#) suggest that up to 30 percent of India's total geographical land area has degraded—almost half of this land constitutes agricultural land, particularly unirrigated/rainfed agricultural land.
- Soil comprises organic matter that consists of elements such as carbon, nitrogen, and phosphorus. Approximately [50 percent](#) of soil organic matter is made up of carbon. Thus, soil carbon measurement provides a good indication of overall soil health.

- In his [2017 paper](#), Australian microbiologist and climatologist Walter Jehne said that the earth's soil carbon sponge, which is formed by the mixture of mineral and organic detritus and air, has an enhanced capacity to absorb and retain rain, improves access to essential nutrients, and supports a diverse range of microbial processes.
- Therefore, when soil carbon levels drop, the ability of the soil to absorb and retain water also reduces. This constrains the effect of rainfall on the soil and leaves the topsoil prone to run-off and erosion, further leading to cycles of floods and droughts. T
- Through carbon sequestration, we can increase soil carbon levels and reduce carbon dioxide levels, which acts as a dual benefit to climate change and agriculture.
- Soil carbon sequestration is a process by which the air's carbon dioxide is extracted and stored in a soil carbon pool. During photosynthesis, plants break down carbon dioxide and water from the atmosphere into oxygen, sugar, and carbon-rich compounds.
- These reach the roots and the soil underneath and nourish the organisms below the soil.

Carbon credit monetisation can be the nudge

- Since the improvement of soil health is innately linked with the ability to increase soil carbon levels, achieving it demands the continuous monitoring of soil carbon levels and the incentivisation of its improvement.
- Monetising soil carbon requires a good understanding of carbon credits. Carbon credits are certificates that represent quantities of greenhouse gases that have been kept out of the air or removed from it.
- One carbon credit certifies that one metric tonne of carbon dioxide has been removed from the atmosphere.
- Advancements in remote sensing data and AI have enabled the prediction of carbon levels through satellite data, and this serves as one of the methods through which carbon credits are calculated.
- Companies and governments purchase carbon credits to meet the climate commitments that they make.

Farmers can benefit from participating in the carbon market

- The direct benefit is that farmers receive cash-based incentives for the carbon they have helped sequester in their lands.
- A farmer who sequesters one carbon credit can earn approximately INR 780 at current market prices, but large corporations are likely to provide better rates—as high as INR 2,000—to farmers when directly purchasing large chunks of carbon credits. In our experience, farmers who follow regenerative practices are able to sequester one to four carbon credits per acre.
- The indirect benefit that farmers experience is the improvement in soil health due to the carbon captured in the soil.
- This improvement can be gauged by determining whether the soil displays any of the following characteristics: Increased water-holding capacity, lower soil density, increased water infiltration, increased nutrient availability, and decreased soil surface temperature.

Carbon storage research

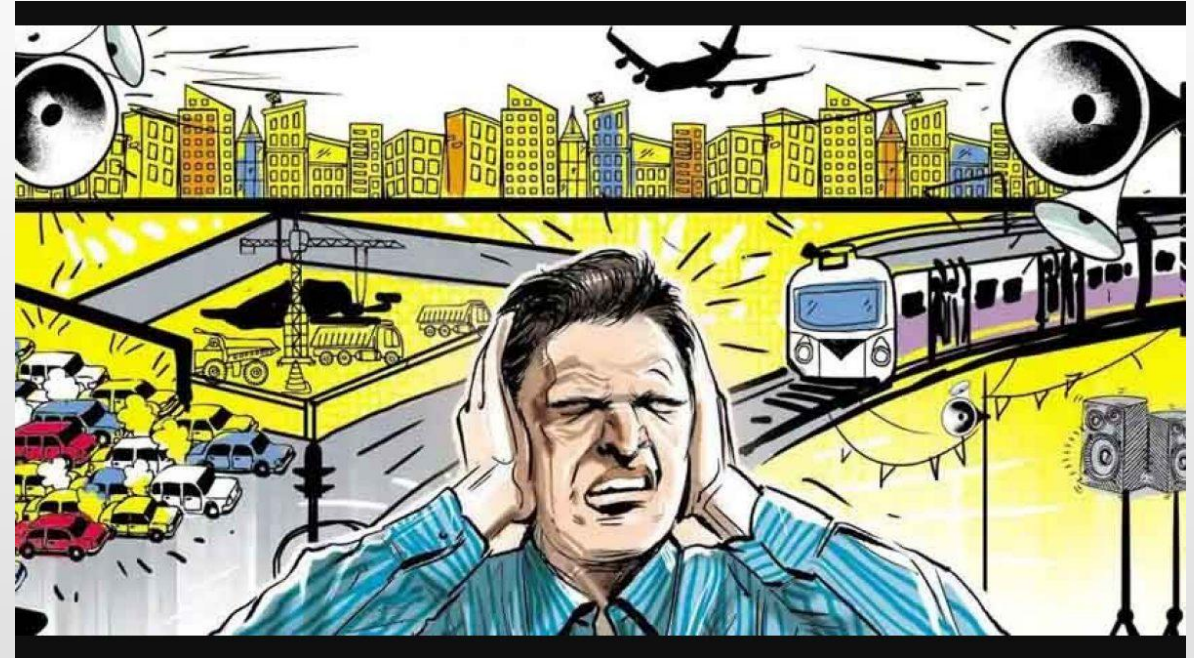


- <https://www.energy.gov/fecm/carbon-storage-research>
- https://www.e3s-conferences.org/articles/e3sconf/pdf/2020/40/e3sconf_te-re-rd2020_03021.pdf
- <https://www.frontiersin.org/articles/10.3389/fclim.2019.00008/full>
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Noise Pollution



➤ **noise pollution**, unwanted or excessive sound that can have deleterious effects on human health, wildlife, and environmental quality. Noise pollution is commonly generated inside many industrial facilities and some other workplaces, but it also comes from highway, railway, and airplane traffic and from outdoor construction activities.



<https://www.onmanorama.com/news/kerala/2019/04/12/noise-pollution-vicinity-dial-this-number-remedy.html>

- Noise pollution is the pollution caused by sound which results in various problems for Humans.
- A sound is a form of energy that enables us to hear.
- We hear the sound from the frequency range of 20 to 20000 Hertz (20kHz). Humans have a fixed range for which comfortably hear a sound if we are exposed to loud sound above our hearing range for quite some time it can permanently damage our hearing capacity.
- In simple language, we can say that any unwanted and undesirable sound is noise.
- It can be anything for example; loud music which we don't want to hear or the blowing of a horn at the traffic signals.
- So, we can define **Noise Pollution** as the presence of unwanted, undesirable, and unnecessary sounds in our environment.

- Sound waves are vibrations of air molecules carried from a noise source to the ear.
- Sound is typically described in terms of the loudness (amplitude) and the pitch (frequency) of the wave.
- Loudness (also called sound pressure level, or SPL) is measured in decibels (dB).
- The normal human ear can detect sounds that range between 0 dB (hearing threshold) and about 140 dB, with sounds between 120dB and 140 dB causing pain (pain threshold).
- The ambient SPL in a library is about 35 dB, while that inside a moving bus or subway train is roughly 85 dB; building construction activities can generate SPLs as high as 105 dB at the source.
- SPLs decrease with distance from the source.

Sound Pressure Level

Sound pressure is the pressure applied by the sound waves. The sound pressure level is the ratio of the actual sound pressure and the reference level which is the lowest intensity sound that can be heard by maximum people. In other words, a sound pressure level decides the amount of pressure on the sound. It provides the logarithmic measures of rms value of the sound pressure to the reference value of the sound. It is expressed in (dB).

The sound pressure level formula is expressed by

$$L_p = 20 \log_{10}(p_{rms} / p_{ref})$$

where, p_{rms} = root mean square sound pressure in Pa,
 p_{ref} = reference sound pressure in Pa. ($20 \mu Pa$)

Solved Example

Example 1: Determine the sound pressure level if the rms pressure value is given as $30 \mu\text{P}_a$.

Solution:

Given

$$P_{\text{rms}} = 30 \mu\text{P}_a$$

$$p_{\text{ref}} = 20 \mu\text{P}_a$$

The sound pressure level formula is given by

$$L_p = 20 \log_{10} (p_{\text{rms}} / p_{\text{ref}})$$

$$L_p = 20 \log_{10} (30 / 20)$$

$$L_p = 3.52 \text{ dB}$$

Example 2: Calculate the sound pressure level in decibel for a given sound pressure of 20×10^3 Micro-Pascal.

The sound pressure level in decibels is calculated by : $L_p = 20 \times \log_{10} \left(\frac{p_{\text{rms}}}{p_{\text{rms}_0}} \right)$

Given, sound pressure (p_{rms}) = 20×10^3 Micro-pascal = 0.02 pascal

Reference sound pressure (p_{rms_0}) = 20 micro pascal = 20×10^{-6} pascal

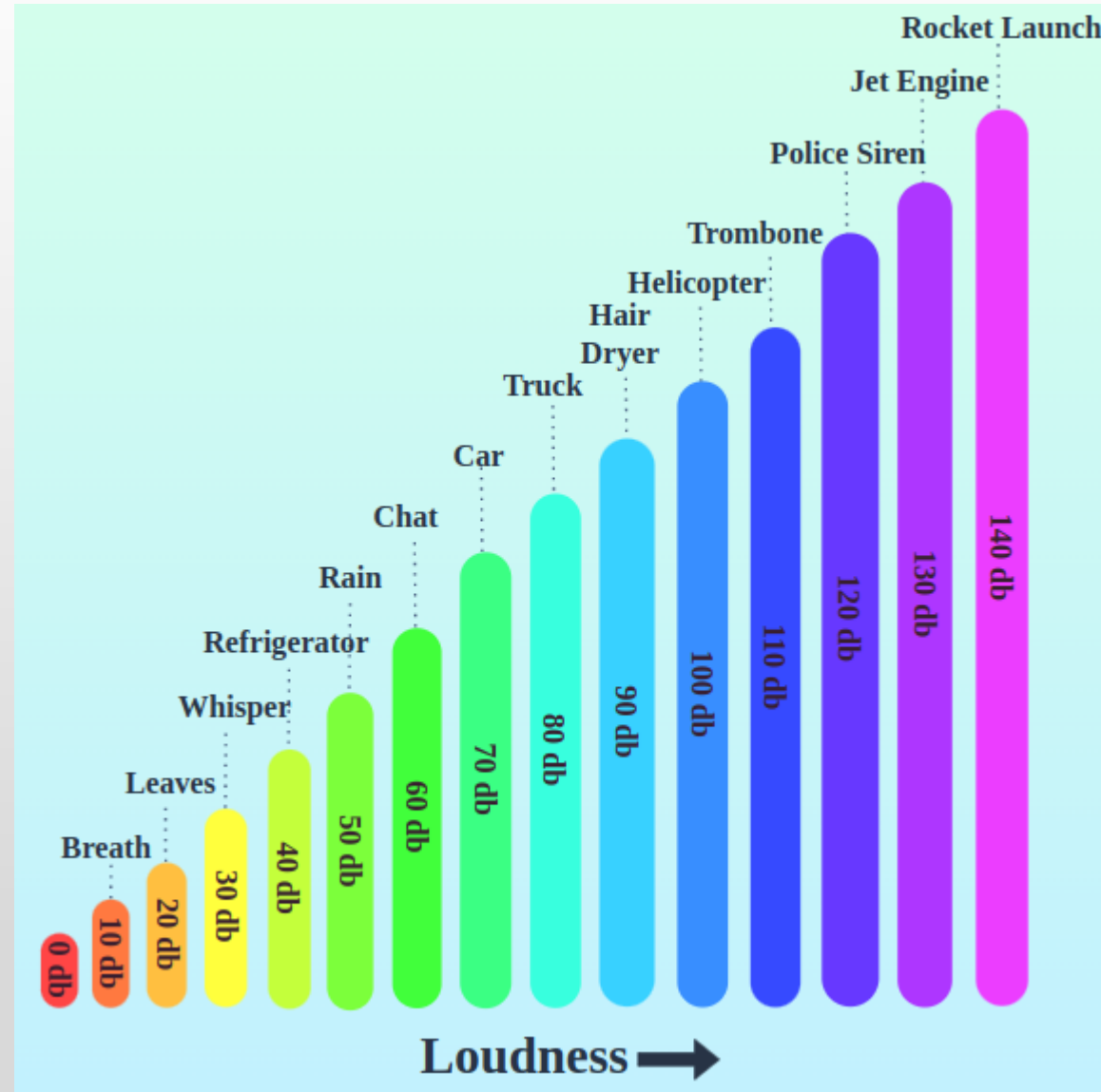
∴ The sound **pressure level in decibels**

$$L_p = 20 \times \log_{10} \left(\frac{p_{\text{rms}}}{p_{\text{rms}_0}} \right) = 20 \times \log_{10} \left(\frac{0.02}{20 \times 10^{-6}} \right) = 20 \times \log_{10}(10^3) L_p = 20 \times 3 = 60$$

Measuring and perceiving noise pollution



- We can identify sound waves as disturbances in the air molecules around us these waves propagate from the source to our ears and when we feel these waves we process them as sound waves.
- Sound can not travel in a vacuum it always needs some material medium to propagate.
- The intensity of the sound is measured with respect to the amplitude of the sound wave. The more the amplitude the more is the intensity of the sound wave.
- Another important factor of the sound wave is its pitch which is described by its frequency.
- The Loudness of the sound is another important characteristic of the sound wave.
- We measure the loudness of the sound in decibels or dB. The normal can hear a sound ranging from 0 dB to 140 dB, (sound in the range from 120 dB to 140 dB causes pain to humans.)
- The noise level in a Library is close to 30 -35 dB while in a moving Bus or Train it is 80-85 dB. The intensity of the sound decreases drastically with respect to the distance between the source and the observer.



<https://www.geeksforgeeks.org/noise-pollution/>

Man-Made Noise

- All the noises which are generated due to human activities are referred to as Man-Made Noises. For example, construction work, noise from automobiles, household appliances, etc.
- This type of noise can range between 30dB to 140dB which is very much harmful.

Environmental Noise

- All the noises that are generated due to environmental activities. For example, the sound produced by animals, the sound of thunderstorms, cyclones, hurricanes, etc.
- Environmental noise can often go up to 140dB which is again a very harmful number for human mankind.

Sources of noise pollution



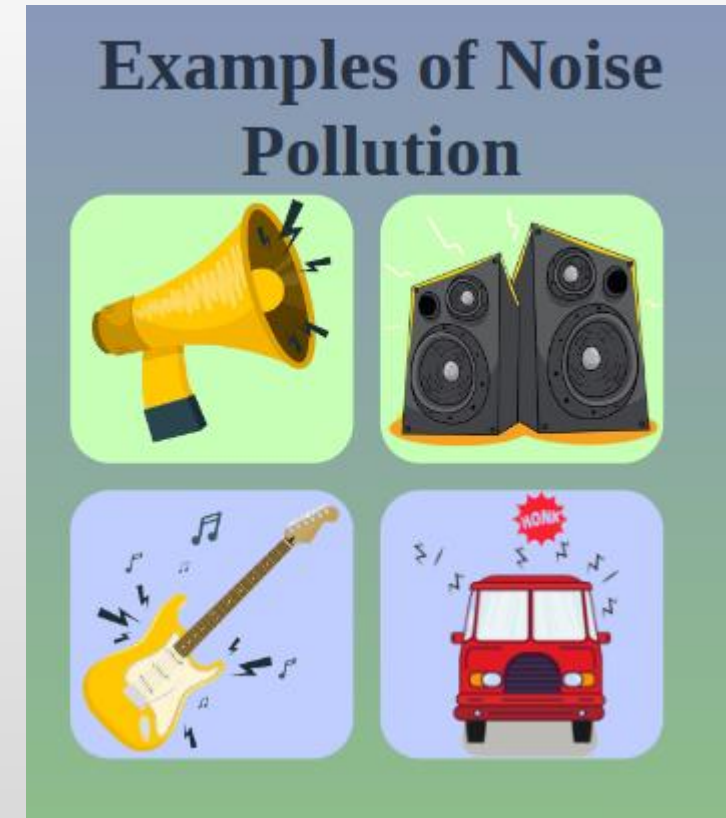
- Bursting of crackers on various social events or religious functions.
 - The loud sound of Radio, Speaker Systems, and Television is also a major source of Noise Pollution.
 - Construction Activities such as constructing buildings or any real estate architecture.
 - Industries are also major sources of Noise Pollution because the machines they use produce noise in an excessive amount which is unbearable.
 - Automobiles also generate a lot of noise pollution in the surrounding by the sound of their engine and by blowing horns excessively.
 - The aviation field is another major source of Noise Pollution as the taking-off and landing of a plane make undesirable and unbearable noise up to 130dB.
 - Home appliances such as Mixer grinders, Washing Machines, Pressure cookers, etc. also produce unwanted sounds i.e. Noise.
 - Animal sounds such as the howling or the barking of a dog are also a source of Noise Pollution.
- Sound of lightning and thunderstorm, Hurricanes, and cyclones also cause noise pollution.

Noise pollution examples



Various examples where noise pollution is observed in our daily life are,

- The honking of the vehicles causes noise pollution.
- Various Industries using heavy machinery cause noise pollution.
- The unnecessary use of loudspeakers used for functions, rallies, and religious purposes causes noise pollution.
- Fireworks used for prolonged periods of time cause noise pollution.
- Loud noise from the construction side cause noise pollution.
- Loud noise from aircraft, jet engines, and railway engines also causes noise pollution.



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Effects of noise pollution

Noise pollution has various effects and some of them are,

Hearing Problems: Loud noise can cause great harm to our ears i.e. reduces the hearing capability of our ears.

Physical Issues: Noise Pollution can cause headaches, high blood pressure, and heart attacks.

Psychological Issues: It can cause hypertension, depression, anxiety, fatigue, and stress in humans as well as animals.

Sleeping Disorders: Excessive noise disturbs our sleeping patterns which may lead to extreme health issues.

Loss of Concentration: Noise distracts the person's mind and a person finds it difficult to concentrate or focus on work or study.

Effect on Wildlife: Wild animals also get disturbed heavily by the noises which humans make, they fear out, faint and in some cases die.

Prevention of noise pollution



- We should avoid playing speakers, radios, and television loudly at home.
- The bursting of crackers unnecessarily should be avoided.
- The horns of vehicles should not be blown unnecessarily and vehicles should be serviced from time to time to prevent the noisy engine.
- Trees should be planted around the buildings and along the roads to reduce noise from roads.
- Sources of noise such as industries, airports, etc., and residential areas should be separated.
- Fine for exceeding certain noise limits.
- Keep checking the surrounding noise levels and limit the sounds that you produce.

Thank You

Name of the Faculty: Dr.Gulmi Chakraborty

Department of Chemistry

Mail Id: gulmi.chakraborty@cgu-odisha.ac.in

www.cgu-odisha.ac.in



C. V. Raman
Global University
ODISHA BHUBANESWAR INDIA